

MATTHEW MERIOLES

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EXPERIENCE

Qualcomm

Senior Engineer (Embedded Linux & Networking)

Jul 2026 – Present

- Design, develop, and execute system-level test plans for QCS compute platforms, including Linux-based operating systems such as Ubuntu, OpenWRT, and Linux LE.
- Collaborate with cross-functional teams (software, hardware, and systems engineering) to define integration strategies and resolve platform-level issues.
- Perform integration and validation testing to ensure compute platform functionality, performance, and reliability.

Northrop Grumman

Electrical Engineer (Network Infrastructure & Automation)

Nov 2025 – Jul 2026

- Produced and owned network infrastructure automation workflows using Python, Ansible, and Jinja2 to generate, validate, and deploy distributed configurations across 300+ devices.
- Troubleshoot EVPN/VXLAN data center fabric issues across multiple production environments, integrating Palo Alto firewalls and VMware NSX-T network virtualization.
- Built CI/CD pipelines using Docker, pytest, and Batfish to package, version, and test network automation tooling.

Associate Electrical Engineer (Network Observability)

Sep 2024 – Nov 2025

- Engineered network observability and telemetry pipelines using Telegraf, InfluxDB, and Grafana; integrated open-source tooling such as LibreNMS, Oxidized, and SuzieQ to monitor infrastructure state and configuration drift.
- Delivered deployment and operations documentation using reStructuredText, Sphinx, and Jinja2, reducing deployment-related support requests and enabling consistent cross-team handoff.

Electronics & Payloads Intern (Validation Automation)

Jun 2023 – Aug 2023

- Developed modular Python tooling built on Paramiko to automate configuration auditing across distributed network devices, reducing a 5+ hour compliance validation process to minutes.
- Collaborated with network engineers to deploy and troubleshoot enterprise network infrastructure, including Cisco Catalyst 3850 switches and KG-250X HAIPES.

Electronics & Payloads Intern (Network Modeling)

Jun 2022 – Aug 2022

- Used Cisco Modeling Labs to emulate and validate network topologies for integration planning and training purposes.
- Managed Teamcenter and Crucible document review workflows, coordinating design approvals and promoting updated engineering drawings and documentation to production release.

Boeing

Electrical Engineering Intern (Failure Analysis)

Jun 2019 – Aug 2019

- Wrote LabVIEW scripts to test electronic components for performance and reliability in extreme environments.
- Recorded and submitted component test data to support product development and engineering decisions.

PROJECTS

802.11 OFDM PHY Implementation

- Implemented a Python-based IEEE 802.11 OFDM baseband chain including STF/LTF preambles, IFFT/FFT processing, cyclic prefix insertion, QPSK modulation, and channel impairment simulation.

GPU-Accelerated Elliptic Curve Cryptography (ECC) Framework

- Built an elliptic curve cryptography framework in Python/CUDA supporting ECDH key exchange and GPU-accelerated scalar multiplication, with automated pytest validation, performance benchmarking, and CI/CD regression testing.

SKILLS

Networking: TCP/IP · BGP · EVPN/VXLAN · VMware NSX-T · NX-OS · IOS-XE · PAN-OS

Languages: Python · Ansible · Jinja2 · C/C++ · Bash · MATLAB

Tooling: Linux (RHEL) · Docker · GitLab CI/CD · Grafana · InfluxDB · Telegraf · Git

Spoken: English (Native) · Japanese (Professional Working Fluency) · Spanish (Basic)

Certifications: Active TS/SCI Security Clearance · Security+

EDUCATION

University of California, San Diego

M.S. Electrical Engineering, Communication Theory & Systems

Expected Jun 2027

B.S. Electrical Engineering, Machine Learning & Controls

Jun 2024